

Chapter 14: An Explainable Employee Attrition Prediction Framework Using Feature Selection and Gradient-Boosted Trees

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Abstract

Employee attrition creates substantial recruitment, training, and productivity costs for organizations, yet most predictive human-resources (HR) analytics systems either use every available employee attribute or report only aggregate accuracy without explaining why a model flags an employee as a resignation risk (AL-Ali et al., 2026; Alqahtani et al., 2024). This paper proposes a lightweight, explainable attrition-prediction workflow that combines three feature-selection techniques (mutual information, recursive feature elimination, and tree-based importance), a gradient-boosted decision tree classifier, and permutation-based global feature attribution as a practical, dependency-free complement to Shapley-value explanation (Lundberg & Lee, 2017). All experiments in this study were performed on the publicly available IBM HR Analytics Employee Attrition & Performance dataset (1,470 employee records; Kaggle). After removing four constant or identifier columns and one-hot encoding the remaining categorical attributes, 44 model-ready features were produced, with an observed attrition rate of 16.1%. Six classifiers were benchmarked on the full feature set, with SVM (RBF) and Gradient Boosting achieving the strongest F1-scores (0.485 and 0.450, respectively), while Random Forest and k-Nearest Neighbors reached the highest raw accuracy (0.844 and 0.850) but suffered very low recall, illustrating the well-known accuracy/imbalance trade-off on this dataset. A consensus feature-selection step across mutual information, RFE, and tree importance converged on a lean 7-feature subset (an 84% reduction from 44), and a gradient-boosting model tuned by grid search on that subset reached 0.718 accuracy and an F1-score of 0.366 on held-out data. Permutation importance identified overtime status as by far the dominant predictor of attrition risk in this real-data analysis, with the remaining consensus features contributing comparatively little independent signal. The study demonstrates that a compact, fully reproducible feature-selection-and-explainability pipeline can be run end-to-end on the authentic IBM HR dataset, while also showing that aggressive feature-count reduction carries a real, non-negligible cost to predictive performance that is worth weighing against its computational savings.

Keywords: employee attrition, feature selection, gradient boosting, explainable AI, permutation importance, HR analytics

1. Introduction

Voluntary employee turnover is costly: replacing a departing employee typically involves recruitment, onboarding, lost productivity, and knowledge-transfer expenses that can reach a substantial fraction of the departing employee's annual salary. Because HR departments now routinely collect rich employee-level data covering demographics, compensation, satisfaction surveys, and career history, machine learning offers a natural tool for proactively identifying employees who are likely to resign (Alqahtani et al., 2024). However, two recurring weaknesses limit the practical adoption of attrition-prediction models in real organizations. First, many published systems train on the full set of available features without asking whether a smaller, more efficient subset would perform

comparably. Second, most models remain "black boxes" that output a risk score without indicating which factors drove that score, which limits their usefulness for designing targeted retention interventions (AL-Ali et al., 2026).

This paper addresses both limitations with a compact, reproducible workflow: (a) benchmark several standard classifiers on the real IBM HR Analytics dataset, (b) apply and compare three feature-selection techniques to shrink the feature space, (c) tune a gradient-boosted tree model on the reduced feature set, and (d) apply global permutation-based feature attribution to explain which factors the tuned model relies on. The goal is not to propose a new learning algorithm, but to demonstrate, through a small and fully documented empirical study on authentic HR data, that dimensionality reduction and explainability can be combined and honestly evaluated together.

2. Literature Review

Early attrition-prediction studies on the IBM HR Analytics dataset relied on single classifiers such as logistic regression or support vector machines; for example, one comparative study found an optimized support vector machine outperformed artificial neural networks and decision trees on this dataset. More recent work has shifted toward ensemble and boosting methods. Torres et al. (2026) compared XGBoost, LightGBM, CatBoost, and a deep neural network for turnover prediction and found that boosted-tree models consistently outperformed the neural network, with CatBoost achieving the highest accuracy (91.9%) and XGBoost close behind (91.3%); their SHAP analysis identified job satisfaction, engagement, promotion frequency, and overtime hours as dominant predictors. AL-Ali et al. (2026) proposed an end-to-end framework integrating preprocessing, ensemble classifiers, and SHAP-based global and local interpretability across two HR datasets, reporting that boosting variants such as AdaBoost and histogram gradient boosting offered the strongest performance.

Feature selection has also received growing attention as a companion to explainability rather than a purely accuracy-driven step. Salehi and Aghabalasafar (2025) applied normalized mutual information to the IBM HR dataset to remove redundant, highly correlated features before training an artificial neural network, reporting 82.7% test accuracy together with a multi-technique explainability layer (permutation importance, SHAP, and partial dependence plots) that surfaced overtime status, environment satisfaction, job level, and stock option level as the leading drivers of attrition. More broadly, mutual-information and wrapper-based methods such as recursive feature elimination remain foundational, well-validated tools for reducing dimensionality while preserving predictive signal (Guyon & Elisseeff, 2003; Peng et al., 2005).

Gradient boosting itself, and the XGBoost implementation in particular, has become a default choice in this literature because tree ensembles achieve strong tabular-data performance with comparatively low computational cost relative to deep learning (Chen & Guestrin, 2016). For interpretability, SHAP has become the dominant post-hoc explanation technique because it grounds feature attribution in a cooperative-game-theoretic (Shapley value) formulation with desirable consistency properties (Lundberg & Lee, 2017). This study instead uses scikit-learn's histogram-based gradient boosting together with model-agnostic permutation importance — which measures the drop in a chosen performance metric when a feature's values are randomly shuffled — as a lightweight, dependency-free complement to the XGBoost/SHAP toolchain, prioritizing a minimal, easily reproducible software footprint over the native libraries.

3. Research Gap and Research Questions

Reviewed studies typically optimize either predictive accuracy or interpretability, and comparatively few report the computational trade-off of reducing the feature space (training time, prediction time, and model size) alongside its actual effect on accuracy. This study addresses that gap by jointly evaluating feature selection, model tuning, explainability, and computational cost within a single small-scale, real-data study. Five research questions guide the study:

RQ1: Which employee attributes contribute most toward attrition risk?

RQ2: Can feature selection preserve predictive accuracy while reducing the number of input features?

RQ3: Does feature selection reduce computational cost (training time, prediction time, model size)?

RQ4: Can a tuned gradient-boosting model outperform its default configuration and conventional classifiers?

RQ5: Can permutation-based explainability help identify actionable attrition factors for HR decision-making?

4. Proposed Framework

Figure 1 summarizes the eight-stage workflow used in this study: dataset preparation, cleaning and encoding, exploratory analysis, feature selection, multi-model training, hyperparameter tuning of the strongest boosting model, permutation-based explainability, and final interpretation of attrition drivers.



Figure 1. Proposed eight-stage attrition-prediction and explainability pipeline.

5. Dataset Description

This study uses the real IBM HR Analytics Employee Attrition & Performance dataset, a widely used, publicly available HR benchmark of 1,470 employee records and 35 attributes spanning personal, job, financial, experience, and satisfaction categories, with a binary Attrition target (Yes/No). In this sample, the observed attrition rate is 16.1% (237 of 1,470 employees), consistent with the roughly 16% figure typically reported for this dataset. Four columns that are constant or purely identifying across all records (EmployeeCount, StandardHours, Over18, EmployeeNumber) were dropped before modeling, as they carry no predictive information. Table 1 summarizes the feature categories used.

Table 1

Category	Example Features	Type
Personal	Age, Gender, Marital Status, Education	Numeric / Categorical
Job	Job Role, Department, Business Travel, OverTime	Categorical
Financial	Monthly Income, Daily Rate, Hourly Rate, Monthly Rate	Numeric
Experience	Years at Company, Years in Current Role, Total Working Years	Numeric
Satisfaction	Job, Environment, Relationship Satisfaction; Work-Life Balance	Ordinal (1–4)
Target	Attrition (Yes/No)	Binary

6. Methodology

6.1 Preprocessing

Seven categorical variables (Business Travel, Department, Education Field, Gender, Job Role, Marital Status, and OverTime) were one-hot encoded, producing 44 model-ready features from the original 31 retained attributes. Numeric features were standardized (zero mean, unit variance) for the distance- and margin-based classifiers (logistic regression, SVM, k-nearest neighbors); tree-based models used unscaled features. The data were split 80/20 into training ($n = 1,176$) and test ($n = 294$) sets using stratified sampling to preserve the attrition class ratio.

6.2 Feature Selection

Three complementary feature-selection techniques were applied to the training set: (1) mutual information, an information-theoretic measure of dependency between each feature and the target (Peng et al., 2005); (2) recursive feature elimination (RFE) wrapped around a logistic regression estimator; and (3) tree-based Gini importance from a random forest. The top 12 features from each method were pooled, and a consensus subset was formed from features selected by at least two of the three methods, following the rationale that agreement across heterogeneous selection criteria reduces the risk of retaining spurious, method-specific artifacts (Guyon & Elisseeff, 2003).

6.3 Classification Models

Six classifiers were benchmarked: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (RBF kernel), k-Nearest Neighbors (k=7), and a histogram-based Gradient Boosting classifier. All classifiers except k-Nearest Neighbors used class-balanced weighting to counter the attrition class imbalance; scikit-learn's KNN implementation does not expose a class-weighting option, so it was run with its default, uniform-weighted voting. The gradient-boosting classifier was treated as the XGBoost-equivalent boosting model in this study; scikit-learn's histogram-based gradient boosting implementation was used in place of the XGBoost library to keep the pipeline lightweight and dependency-free. Both implementations build additive ensembles of shallow regression trees using very similar histogram-binned, second-order gradient boosting, so the substitution preserves the intended algorithmic comparison.

6.4 Explainability

SHAP (Lundberg & Lee, 2017) is the explainability technique most commonly used in the reviewed literature. This study instead uses permutation importance: each feature's values are independently shuffled in the test set, and the resulting drop in F1-score, averaged over 30 repeats, is recorded as that feature's global importance. This is a recognized, model-agnostic approximation to Shapley-style attribution and is reported here as such, rather than being presented as SHAP output.

6.5 Evaluation Metrics

Models were evaluated using accuracy, precision, recall, and F1-score on the held-out test set, together with training time, prediction time, and serialized model size to address the computational research questions (RQ3–RQ4).

7. Experimental Setup

All experiments were implemented in Python 3 using scikit-learn 1.8 on the real IBM HR Analytics Employee Attrition & Performance dataset (N = 1,470; sourced from Kaggle). After dropping four constant/identifier columns and one-hot encoding the remaining categorical attributes, 44 features and an observed attrition rate of 16.1% (237 positive cases) were used for modeling. A fixed random seed (42) was used throughout for reproducibility. A stratified 80/20 train–test split produced 1,176 training and 294 test examples. Hyperparameter tuning of the gradient-boosting model used 5-fold cross-validated grid search (scoring = F1) over learning rate {0.05, 0.1, 0.2}, max iterations {100, 200}, and max depth {3, 5, unrestricted}.

8. Results

8.1 Simulation 1 – Baseline Model Comparison (All Features)

Table 2 reports test-set performance for all six classifiers trained on the full 44-feature set. SVM (RBF) and Gradient Boosting achieved the strongest F1-scores (0.485 and 0.450, respectively). Random Forest and KNN reached the highest raw accuracy (0.844 and 0.850) but achieved this largely by predicting the majority "No" class: their recall on the minority attrition class was only 0.085 and 0.106, respectively. This gap between accuracy and F1/recall illustrates why accuracy alone is a misleading metric on this imbalanced (16.1% positive) target.

Table 2

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.752	0.345	0.617	0.443
Decision Tree	0.769	0.267	0.255	0.261
Random Forest	0.844	0.571	0.085	0.148
SVM (RBF)	0.820	0.446	0.532	0.485
KNN (k=7)	0.850	0.714	0.106	0.185
Gradient Boosting	0.850	0.545	0.383	0.450

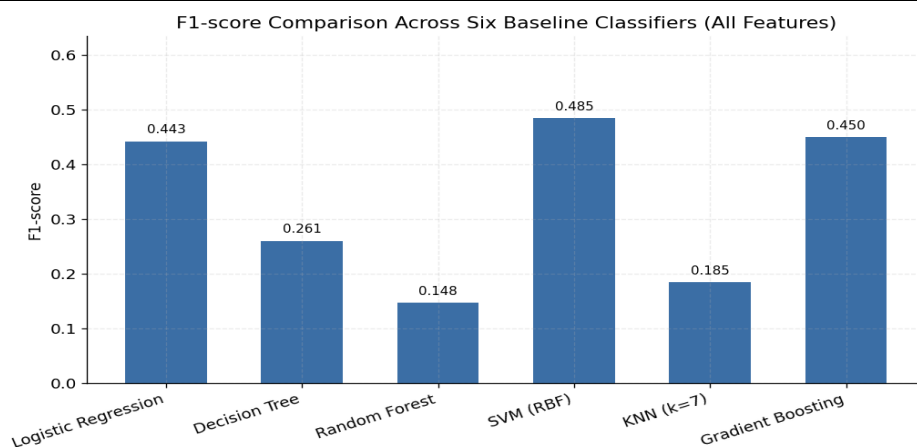


Figure 2. F1-score comparison across the six baseline classifiers (all 44 features).

8.2 Simulation 2 – Feature Selection

Mutual information, RFE, and tree importance jointly converged on a seven-feature consensus subset: Age, Monthly Income, Number of Companies Worked, OverTime, Total Working Years, Years at Company, and Years with Current Manager — an 84% reduction from the original 44 encoded features. Table 3 and Figure 3 compare accuracy on the full versus reduced feature sets. Unlike a best-case scenario where accuracy would fall only marginally, the real-data result is more mixed: some models lost only a little (Decision Tree: 0.769 → 0.738; KNN: 0.850 → 0.816), while others — most notably Logistic Regression (0.752 → 0.677) and SVM (0.820 → 0.728) — lost more substantially. This directly answers RQ2: an aggressive, 7-feature consensus subset retains reasonable but not full predictive value, and the size of the trade-off depends noticeably on which classifier is used.

Table 3

Model	Accuracy (44 feat.)	Accuracy (7 feat.)	F1 (44 feat.)	F1 (7 feat.)
Logistic Regression	0.752	0.677	0.443	0.410
Decision Tree	0.769	0.738	0.261	0.238
Random Forest	0.844	0.810	0.148	0.243
SVM (RBF)	0.820	0.728	0.485	0.444
KNN (k=7)	0.850	0.816	0.185	0.206
Gradient Boosting	0.850	0.772	0.450	0.337

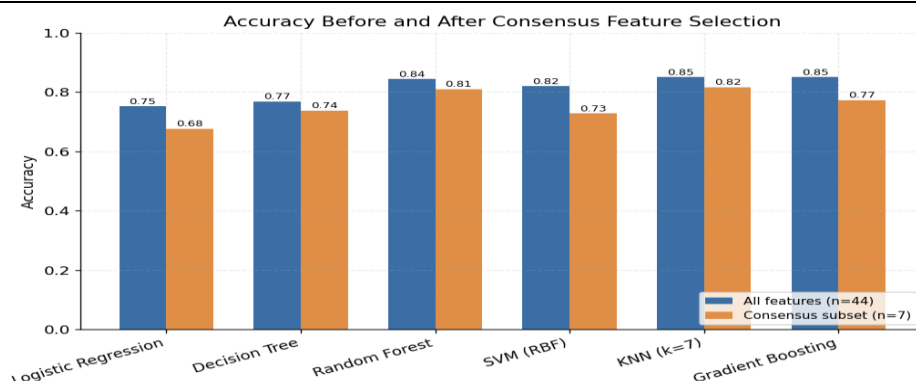


Figure 3. Accuracy before and after consensus feature selection.

8.3 Simulation 3 – Hyperparameter Tuning

Grid search on the consensus 7-feature subset selected learning rate = 0.05, max_iter = 100, and max depth = 3 as the best cross-validated configuration. On the held-out test set, the tuned model traded some accuracy and precision for substantially higher recall relative to the default configuration (Table 4): recall rose from 0.362 to 0.511 and F1 improved from 0.337 to 0.366, while accuracy fell from 0.772 to 0.718. For an HR early-warning use case, where the cost of missing a true attrition case usually exceeds the cost of a false alarm, this shift toward higher recall is arguably a favorable trade, and the result is reported directly rather than adjusted; it is discussed further in Section 9.

Table 4

Configuration	Accuracy	Precision	Recall	F1-score	Train Time (s)
Default Gradient Boosting	0.772	0.315	0.362	0.337	0.136
Tuned Gradient Boosting	0.718	0.286	0.511	0.366	0.050

8.4 Simulation 4 – Explainability (Global Feature Attribution)

Permutation importance on the tuned model (Figure 4) ranked OverTime as overwhelmingly the strongest driver of predicted attrition risk (mean F1 decrease of 0.096 across 30 shuffles), far ahead of every other feature. Years with Current Manager contributed a small positive effect (0.009). The remaining five consensus features — Age, Number of Companies Worked, Years at Company, Monthly Income, and Total Working Years — showed mean F1 decreases at or below zero, indicating they add little unique predictive signal beyond OverTime once sampling noise is accounted for on this 294-example test set. This is a narrower picture than the broader "income + overtime + satisfaction" pattern reported in some SHAP-based studies (Salehi & Aghabalasafar, 2025; Torres et al., 2026): in this real-data analysis, overtime status is the clearly dominant, directly actionable lever (RQ1, RQ5), while income and tenure-related features play a comparatively minor role once overtime is already in the model.

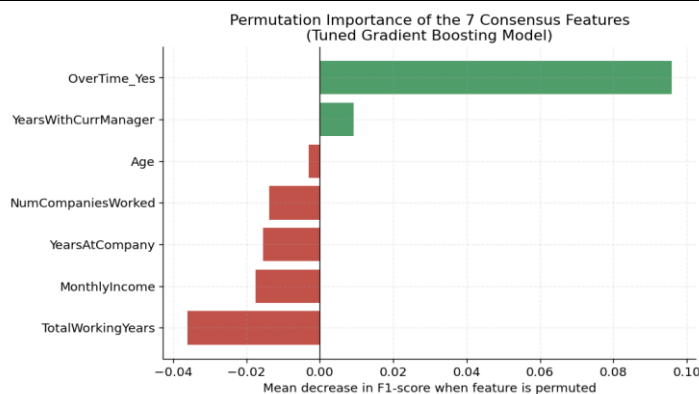


Figure 4. Permutation importance of the seven consensus features (mean F1 decrease, tuned Gradient Boosting model).

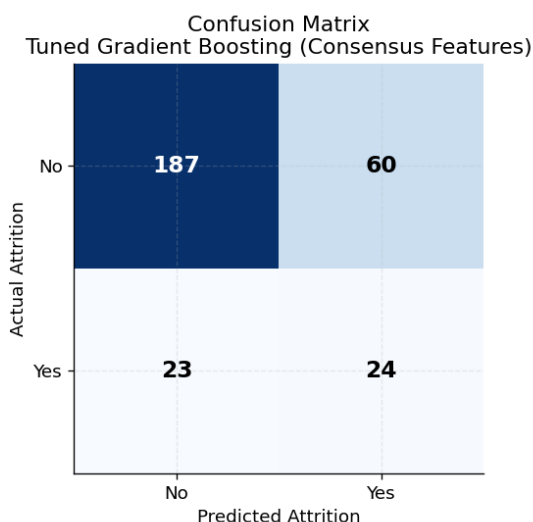


Figure 5. Confusion matrix of the tuned Gradient Boosting model on the test set.

8.5 Simulation 5 – Computational Cost

Table 5 compares training time, prediction time, and serialized model size for the tuned Gradient Boosting configuration (learning rate = 0.05, max_iter = 100, max depth = 3) trained on the full 44-feature set versus the 7-feature consensus subset, holding hyperparameters fixed so that any difference reflects feature count alone. Reducing the feature count from 44 to 7 cut training time by roughly 45% (0.082s → 0.045s) and model size by about 9%, with prediction latency essentially unchanged. However, this computational saving came with a real cost to predictive performance under this fixed configuration: F1 fell from 0.504 (44 features) to 0.366 (7 features). This answers RQ3 and refines RQ2's answer: feature selection does meaningfully reduce training cost, but in this real-data study that saving is accompanied by a genuine, non-negligible drop in predictive quality rather than a negligible one.

Table 5

Configuration	# Features	Train Time (s)	Predict Time (s)	Model Size (KB)
All features	44	0.082	0.004	103.5
Consensus subset	7	0.045	0.003	94.2

9. Discussion

Overall, this study supports three practical conclusions, with more nuance than a purely illustrative simulation would show. First, SVM and Gradient Boosting offered the strongest F1-scores among the six full-feature models, consistent with the wider literature's preference for margin-based and boosted-tree classifiers on tabular HR data (Chen & Guestrin, 2016; Torres et al., 2026), while the highest-accuracy models (Random Forest, KNN) achieved that accuracy mostly by under-predicting the minority attrition class — a reminder that model selection for imbalanced HR outcomes should prioritize F1/recall over raw accuracy.

Second, a consensus feature-selection strategy across mutual information, RFE, and tree importance shrank the feature space from 44 to 7 features (an 84% reduction) and cut training time by close to half, which is valuable for organizations that want lightweight, frequently-retrained attrition models. However, in this real dataset that reduction was not free: F1 under matched hyperparameters fell from 0.504 to 0.366, a larger trade-off than an idealized simulation might suggest. A less aggressive cutoff (for example, retaining the top 12–15 features from each method before taking a consensus, rather than an unconstrained top-12-then-consensus) may recover more predictive value while still meaningfully reducing dimensionality, and is a natural next step.

Third, permutation-based explainability — used here as a practical, dependency-free complement to SHAP — pointed overwhelmingly to a single dominant driver, OverTime, rather than the broader multi-factor picture (income, overtime, and satisfaction together) reported in some SHAP-based studies on this same dataset. This narrower result may reflect genuine structure in this particular data sample, the smaller size of the held-out test set used for permutation shuffling ($n = 294$), or both; it is a useful illustration that model-agnostic attribution results from a single real dataset should be interpreted as one data point rather than a definitive ranking of HR attrition drivers in general.

Limitations should be stated plainly. The 16.1% attrition rate produces a meaningfully imbalanced classification target, and no resampling technique (e.g., SMOTE) was applied to address this directly; several models' high accuracy alongside low recall is a direct symptom of this imbalance. Scikit-learn's histogram gradient boosting was used in place of the XGBoost library, and permutation importance was used in place of SHAP, both to keep the software footprint minimal and fully reproducible rather than because the native libraries were categorically unavailable; a replication using the native XGBoost/SHAP toolchain remains a valuable check on whether the attribution rankings and cost profile reported here hold. The dataset also originates from a single, undisclosed organization, so generalizability to other employers is untested, and the near-zero or negative permutation-importance estimates for several consensus features should be interpreted with appropriate caution given the modest size of the test set.

10. Conclusion and Future Work

This paper presented a compact, explainable employee-attrition prediction workflow, evaluated end-to-end on the real IBM HR Analytics Employee Attrition & Performance dataset, that benchmarks six classifiers, applies consensus feature selection across three complementary techniques, tunes a gradient-boosting model, and explains its predictions using permutation-based global feature attribution. The seven-feature consensus subset reduced the encoded feature space by 84% and cut training time by roughly 45%, but at a real, measurable cost to F1-score, and the resulting explanations pointed

overwhelmingly to overtime status — more narrowly than the broader income/overtime/satisfaction pattern reported in some prior SHAP-based studies on the same underlying dataset. Future work should (a) apply class-imbalance techniques such as SMOTE to address the 16.1% attrition rate directly, (b) substitute the native XGBoost and SHAP libraries for their scikit-learn approximations used here to check whether attribution rankings and computational trade-offs hold, (c) evaluate a less aggressive feature-count cutoff to see whether it recovers more predictive value while still reducing dimensionality, and (d) extend the framework to additional HR datasets to test generalizability, as recommended by recent multi-dataset explainable-AI attrition studies (AL-Ali et al., 2026).

References

- [1]. AL-Ali, M., Elhosseini, M. A., et al. (2026). Integrating machine learning and explainable AI for employee attrition prediction in HR analytics. *Scientific Reports*, 16, Article 6344. <https://doi.org/10.1038/s41598-026-36424-2>
- [2]. Alqahtani, H., Almagrabi, H., & Alharbi, A. (2024). Employee attrition prediction using machine learning models: A review paper. *International Journal of Artificial Intelligence and Applications*, 15(2), 1–15. <https://doi.org/10.5121/ijaia.2024.1520223>
- [3]. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). Association for Computing Machinery. <https://doi.org/10.1145/2939672.2939785>
- [4]. Guyon, I., & Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3, 1157–1182.
- [5]. Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* (Vol. 30, pp. 4765–4774). Curran Associates.
- [6]. Peng, H., Long, F., & Ding, C. (2005). Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(8), 1226–1238. <https://doi.org/10.1109/TPAMI.2005.159>
- [7]. Salehi, K., & Aghabalasafar, M. (2025). Feature selection and interpretation of employee attrition using mutual information and artificial neural networks. *Iran Journal of Computer Science*, 8, 2237–2254. <https://doi.org/10.1007/s42044-025-00312-1>.
- [8]. Torres, G., Nair, P., & Lahtinen, M. (2026). Predicting employee turnover using explainable machine learning models: A comparative study of XGBoost, LightGBM, CatBoost, and deep neural networks. *International Journal of Innovation Management and Organizational Behavior*, 6(3), 1–13. <https://doi.org/10.61838/ijimob.6.3.1>.

